

Author response (condensed, per-reviewer)

Submission 25189

March 30, 2026

Abstract

We thank the reviewers for their careful reading and constructive feedback. Below we reply **per reviewer**, following the OpenReview structure (“Comment:” / “Response:”), and we indicate concrete camera-ready edits. Each reviewer block is written to fit within typical OpenReview character limits for an author response.

Author response — Reviewer VCqo

Comment:

Some of the figures could be cleaner and clearer. As an example of this, in Figure 2, it is not clear what is shown in Section 2,

you should keep the same citation style throughout the paper. Before line 126 you cite works [1, 2], but in line 139, frozen random features ensure $\text{supp}(W^0(C_1)) \rightarrow$ this is not clear
line 110, second column, what do you mean by the mean-field width limit ? Do you mean the limit as $n \rightarrow \infty$?
Section 3.1.

I understand that the low rank factorization appears from the stacking of W_j^{ℓ} and h_j^{ℓ} in Section 3.2.

Line 166, your notations with the lower and uppercase case c_1 in $W^0(c_1)$ is not really clear. Your expectation is taken with respect to the neurons indices ? which is strange to me ? You should clarify this in Section 3.3.

lines 187-188, I strongly recommend writing down the expression of the loss at least and recalling the context in lines 187-188: What are the learning rate schedules?
Section 3.4.

On lines 166 - 168, you say that the “Better than null” assumption requires the initial loss to be small. Line 178, second column, you say that for Relu, the same holds with high probability in r . I don’t know how easily this can be done but I would recommend laying out the assumptions clearly in Section 3.5 - 3.6

lines 207 - 210, second column, you have $|H_{\ell}(W') - H_{\ell}(W)| < K \|W^{\ell}\|_{\infty}$,
lines 218 - 219, what is the solution operator F ?
lines 220 - 223: “After defining norms, the solution operator $F, \dots$ and the contraction operator Φ ”
Section 3.7

In the statement of Theorem 3.2., when you talk about the mean field limit, you are referring to the limit as $n \rightarrow \infty$.

lines 269 -, What you call “High level proof idea” is pretty opaque. I think I would remove i
 lines 272 - 273, “this yields $\mathbb{E}_Z[\text{upstream} \times \text{local}]$ ” -> what does that mean? I u
 lines 220 - 223: What does $\partial_2 \mathcal{L}$ mean here? Does that mean the partial deriv
 lines 239 - 241, second column, it is not clear to me why the frozen random features yield he
 Section 3.8

In the statement of Theorem 3.3. Here as well, it is not clear why/how you initialize over th
 Same Theorem, what do you mean by the “coupling procedure”?

Same, What does

refer to here ? If I’m not wrong, you don’t introduce it. I understand it measures the devia

Response: We agree and will revise the camera-ready accordingly.

- **Presentation.** Redraw Fig. 2 with a two-color scheme and matched-momentum comparisons; unify citation style; fix minor typos/phrasing.
- **Definitions.** Add explicit loss $\mathcal{L}/\mathcal{L}$, define $\partial_2 \mathcal{L}$, define F (Picard limit / flow map), and clarify the ODE–SGD discretization and learning-rate schedules $\xi(t)$.
- **Frozen features.** Restate the frozen-feature richness assumption in standard terms (random-feature density) and remove ambiguous “supp” wording.
- **Assumptions.** Define “better than null”, define “mixing” ($L^{(\ell)}$, $\|L^{(\ell)}\|_{\infty,1} \leq rK$), and clarify ReLU vs. the nonzero-derivative assumption (GeLU satisfies; ReLU has a high-probability-in- r heuristic under symmetric mixing).
- **Lipschitz bound (explicit).** In our architecture,

$$H_\ell(c_\ell; x, W) = \sum_{k=1}^r L_{c_\ell, k}^{(\ell)} f_{\ell, k}(x; W), \quad \|L^{(\ell)}\|_{\infty,1} := \max_{c_\ell} \sum_{k=1}^r |L_{c_\ell, k}^{(\ell)}|.$$

Hence for two trainable parameter tuples W', W'' (same frozen L),

$$|H_\ell(c_\ell; x, W') - H_\ell(c_\ell; x, W'')| \leq \|L^{(\ell)}\|_{\infty,1} \max_{1 \leq k \leq r} |f_{\ell, k}(x; W') - f_{\ell, k}(x; W'')|.$$

We will replace the ambiguous $\|W^{(\ell)}\|_{\infty,1}$ by $\|L^{(\ell)}\|_{\infty,1}$ and standardize W to denote *trainable* weights only.

- **Solution operator F (explicit).** We use F for the Picard (integral) map of the mean-field ODE: if $\dot{\mu} = \Phi(\mu)$, then

$$(F\mu)(t) = \mu(0) + \int_0^t \Phi(\mu(s)) ds.$$

- **Theorems / proof sketches.** Clarify that Theorem 3.2 is for arbitrary depth in our architecture and specify which widths diverge; shorten/move the high-level proof sketch; replace “upstream $\times$ local” by “backward $\times$ forward”; define “shifts” (neuron-independent translation under i.i.d. collapse in Nguyen & Pham (2023), Theorem 54 / Corollary 30) and explain heterogeneity via frozen mixing (since $H_2(c_2; x, W) = \sum_k L_{c_2, k} f_k(x; W)$ depends on c_2 ; analogous in spirit to bidirectional diversity in Nguyen & Pham (2023), Theorem 38); rewrite Theorem 3.3 initialization language and define D (Wasserstein-type distance as in Nguyen & Pham (2023)).

Author response — Reviewer RZVa

Comment:

While symmetry is preserved at

, it becomes largely imperceptible at

. Since

remains an extremely low-rank constraint relative to a width of

, this sensitivity challenges the practical robustness of the feature learning mechanism .

Theorem 3.3 bounds the finite-width approximation error by

. Does the symmetry loss at
occur because the exponential factor
dominates
, causing divergence from idealized mean-field dynamics? If so, is "symmetry preservation" a
artificially small to prevent mean-field breakdown?

If

must be kept strictly minimal to preserve mean-field behaviors, how does this framework scale
? If evaluated on a synthetic dataset with controlled
, is there a theoretical or empirical phase transition when
? It would be insightful to clarify if the network loses its theoretical structural benefits
exceeds what the practical width
can exponentially suppress.

Response: We thank the reviewer for the detailed and actionable feedback.

- **Scope / “freezing” limitation.** We agree that freezing mixing matrices is not standard end-to-end practice; it is a deliberate architectural choice that makes the deep mean-field analysis tractable while preserving expressive channelized dynamics. We will clarify that our target scope is theory-guided low-rank training (e.g. oscillatory/SciML settings), not a claim of covering all end-to-end deep learning.
- **Novelty vs. Nguyen & Pham (2023).** We will make the novelty explicit: the contribution is not a new mean-field framework but the low-rank architecture + frozen mixing that avoids the deep i.i.d. collapse mechanism and yields a tractable r -channel ODE system, with a global-convergence conclusion within the parametrized class when the dynamics converges.
- **Theorem 4.1 and mechanism.** We agree the two-point toy is minimal; we will add a developed “mechanism” remark after Theorem 4.1 (feedback loop through $H_2 = \sum_k L_{c_2,k} f_k$ and ReLU gates), and we will better separate what is proved from what is heuristic/empirical in higher dimension.
- **Symmetry sensitivity / hyperparameters.** We will add a short variance-over-seeds summary (and, where possible, a table/plot) clarifying which symmetry observable is measured and how it varies with r and optimization settings. We will also adjust claims to avoid over-interpreting a conservative finite-width bound.
- **Exponential Grönwall factor.** The exponential prefactor in Theorem 3.3-type bounds is a worst-case Grönwall artifact; it should not be read as a predictor of practical symmetry breakdown. We will rephrase this part and avoid claiming that “channel specialization avoids the exponential bound” without evidence.
- **(d, r, N) scaling.** We will add discussion and, resources permitting, synthetic sweeps varying intrinsic dimension and rank to probe whether/where a transition occurs.

Author response — Reviewer Z9GA

Comment:

As I mentioned in the Strengths and Weaknesses part, the architecture considered in this paper are trained. Could the authors elaborate more on the technical difficulty?

I wonder how universal is spectral bias discovered in Section 4 for low-rank training? For ex

Response: We thank the reviewer for the clear summary of the technical concerns.

- **Constant rank / one trained factor.** We agree this is not the most general practical setting; it is the setting in which we can (i) avoid deep i.i.d. collapse while (ii) keeping a closed mean-field description and a tractable global-convergence argument. We will clarify that this is a first step toward understanding low-rank optimization mechanisms in deep networks.
- **“Easier than Nguyen & Pham (2023)”.** Our architecture does simplify certain aspects (frozen mixing reduces degrees of freedom), but it introduces a new channelized coupling (the r -channel forward map $H_\ell = \sum_k L_{\cdot,k} f_k$ and corresponding backward signals), which is exactly where the proof differs: the Lipschitz/coupling chain carries $\|L^{(\ell)}\|_{\infty,1}$ and a $\max_k$ over channels. We will highlight the single step where the argument is not verbatim.
- **Training both factors.** Training both factors removes the frozen heterogeneity that breaks permutation symmetry and substantially complicates the coupling estimates (and may reintroduce collapse modes). We will state this obstacle explicitly and position it as open future work.
- **Spectral bias.** We agree Section 4 is currently qualitative. We will (i) tone down claims, (ii) clearly separate NTK/lazy-regime statements (where classical low-frequency bias persists) from the rich feature-learning regime where we observe different behavior, and (iii) add citations for the polyhedral/Fourier intuition (Rahaman et al. 2019; Montúfar; Raghu) while labeling it as heuristic.

Author response — Reviewer ByWV

Comment:

Typo at (040) "...is low rank all we need for global convergence ? Low-rank networks...": The . substantially...".

At (044), it is posed that "can gradient-based training still converge to a global minimizer, , implies that the low-rank structure affects the dynamics of the gradient-based optimization . Note that the low-rank problem is different than the full-rank problem. Therefore, the func

For the Fig.

, how are different momentum rates selected for the corresponding level of rank constraints?

Do low-rank random feature networks differ from the multi-component multilayer neural network

Typo at (137),

should be denoted by vector notation.

Could you clarify the convergence assumptions made by Nguyen & Pham (2023) for multilayer and

Nguyen & Pham, 2023 established the global convergence of multilayer neural networks trained

Response: We thank the reviewer for the concrete suggestions.

- **Experiments.** We will expand the empirical section (additional datasets/sweeps where space permits) and add variance-over-seeds/frozen-feature draws to quantify sensitivity.
- **What is claimed about global minima.** We agree the current wording can be misleading. Our guarantee is *within the low-rank parametrized class*: low-rank and full-rank problems have different feasible sets and need not share global minimizers. We will state this explicitly and avoid implying equality of full-rank and low-rank optima.
- **Momentum selection.** Chosen by a small sweep; we will report the sweep protocol.
- **Zhang et al. (2023).** The architecture matches multi-component networks; our novelty is the mean-field optimization analysis (channel ODEs, global-convergence conclusion within the

parametrized class, and specialization mechanism). Zhang et al. do not analyze optimization.

- **Assumptions in main text.** We will move a compact assumption block (bounded activations/mixing, initialization, loss regularity, convergence-to-limit-point) into the main text with a short discussion.
- **What is new vs. Nguyen & Pham (2023).** We will emphasize the single non-verbatim step: the low-rank channelized forward map introduces $\|L^{(\ell)}\|_{\infty,1}$ and $\max_k$ over channels in the coupling/Lipschitz chain, and frozen mixing breaks the deep i.i.d. collapse mechanism.
- **Nguyen & Pham (2023) convergence assumptions and SGD.** We will clarify that Nguyen & Pham (2023) use a weighted coupled- L^1 transport condition under a coupling π_t (not merely moment convergence), and that our analysis is in the same mean-field/SGD scaling where SGD is a small-step discretization of the limiting ODE (gradient flow).
- **Typos / vector notation.** We will fix the listed typos and notation inconsistencies.

Author response — Reviewer A6kZ

Comment:

1. Convergence guarentees. Are there conditions under which convergence is guaranteed for RF-
2. High-dimension extensions. Does the spike learning mechanism extend to higher dimensions w
3. Sensititivity of randomness of fixed feature maps. How sensitive are the convergence guara

Response: We thank the reviewer for the thorough and constructive critique.

- **“Channel specialization avoids the exponential bound”.** We agree the current wording is unsupported. We will remove or rephrase this as a conservative statement (the bound is worst-case), and we will add empirical evidence if we keep any specialization-related claim.
- **Minimum network size claim.** We will either add a citation/justification or remove the claim from Appendix F.6.
- **Beyond two points/two channels.** We agree the toy theorem is limited. We will add a more developed treatment: a mechanism remark after Theorem 4.1 explaining the reinforcing-loop effect through $H_2 = \sum_k L_{c_2,k} f_k$ and ReLU gates, and we will clarify that the ODE-level mechanism is dimension-agnostic (the toy proof is chosen for tractability).
- **Variance over frozen features.** We will add variance analysis over different draws of frozen random features (finite width).
- **Presentation fixes.** We will fix the ordering in Section 4.1 (present Theorem 4.1 before referencing it) and revise the figure captions (notably Figs. 8–9) for clarity.
- **Convergence guarantees.** Our main theorem is conditional: if the mean-field dynamics converges, the limit is a global minimizer within the parametrized class. Unconditional convergence rates/guarantees for RF-LR remain largely open; we will clarify this and point to the sharpest known partial results (and what is future work).
- **High-dimensional spike mechanism.** The channel feedback mechanism extends in the same form at the ODE level (same algebra of $H_2 = \sum_k L_{c_2,k} f_k$ and channelwise backward signals), but the clean two-point proof does not directly transfer because high-dimensional features can overlap; we will clarify this distinction and add empirical support.
- **Sensitivity to frozen draw.** The theory assumes bounded mixing; performance at finite width can vary with the frozen draw. We will quantify this variance empirically and clarify which parts of the theory depend on boundedness vs. are mean-field heuristics.